Una Joh

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· Research Interests

I study how computational systems represent information **(1)** to support human sensemaking at scale and **(2)** when computational systems themselves become users of that information.

(1) Computational Social Science, Text Mining, Public Opinion, Computational Narrative Analysis
(2) Human-Centered AI, AI Agents and Memory, Personalization in AI Systems

· Education

Ph.D. in Information Science and Technology (ABD) Expected May 2027
Syracuse University Syracuse, NY, USA

Dissertation: *Computational Representations for Corpus Sensemaking: Interpretable Themes and Latent Narratives*

Develops computational methods for analyzing large-scale public discourse through (1) interpretable topic modeling with sparse autoencoders and (2) a framework for discovering latent narratives in social media. Both are applied to U.S. immigration discourse to represent thematic and narrative structures at scale.

Advisor: Dr. Jeff Hemsley; Committee: Dr. Bei Yu, Dr. Joshua Introne, and Dr. Yiqi Li

M.A. in Economics Aug 2022
Sungkyunkwan University (SKKU) Seoul, Korea

Thesis: *Analysis of Real Estate Market Using News-Based Sentiment Index*

Bachelor of Economics Aug 2020
Sungkyunkwan University (SKKU) Seoul, Korea

· Publications and Manuscripts

Refereed Journal Articles

[J3] **Joh, U.**, Khoury, C., Smith, A. O., Duan, Y., & Hemsley, J. (2026). Climate communication in the stormy transition of Twitter to X. *Information, Communication & Society, 29*(9), 2716–2734. https://doi.org/10.1080/1369118X.2025.2590556

[J2] Li, Y., Mitiaeva, M., Deng, X., **Joh, U.**, & Zhang, Y. (2025). Straddling the boundary of misleading information group: Exploring the association between network brokerage and morality expressions. *Journal of Media Psychology: Theories, Methods, and Applications, 37*(5), 323–332. https://doi.org/10.1027/1864-1105/a000482

[J1] Duan, Y., Hemsley, J., Smith, A. O., **Joh, U.**, Gray, L., & Khoury, C. (2024). Curating virality: Exploring curated logics within #BlackLivesMatter on Twitter/X. *Social Media + Society, 10*(2). https://doi.org/10.1177/20563051241242799

Refereed Conference Proceedings

[C7] Li, Y., Yang, A., & **Joh, U.** (2027, forthcoming). Who drives online community fragmentation? Disentangling social influence and network structure in cyberbalkanization. In *Proceedings of the 60th Hawaii International Conference on System Sciences (HICSS 60)*.

[C6] **Joh, U.**, & Yu, B. (2026, forthcoming). Beyond top words: MonoTM for topic modeling with interpretable monosemantic features. In *Findings of the Association for Computational Linguistics: AACL-IJCNLP 2026*. [arXiv]

[C5] **Joh, U.**, Li, Y., & Hemsley, J. (2026). How growing toxicity manifests: A topic trajectory analysis of U.S. immigration discourse on social media. *Proceedings of the International AAAI Conference on Web and Social Media, 20*(1), 1129–1158. https://doi.org/10.1609/icwsm.v20i1.42686

[C4] Smith, A. O., Cousin, C., **Joh, U.**, Khoury, C., Duan, Y., & Hemsley, J. (2025). Towards memetic legitimation of knowledge: Memes and cultural heritage. *Information Research: An International Electronic Journal, 30*(iConf), 618–626. https://doi.org/10.47989/ir30iConf47260

[C3] Smith, A. O., **Joh, U.**, & Hemsley, J. (2025). Does Google dream of electric memes? From human to computational culture. *Information Research: An International Electronic Journal, 30*(iConf), 287–300. https://doi.org/10.47989/ir30iConf47350
Best Full Research Paper Award

[C2] Duan, Y., Khoury, C., **Joh, U.**, Smith, A. O., Cousin, C., & Hemsley, J. (2024). Comparing climate change content and comments across Instagram Reels, TikTok, and YouTube Shorts and long videos. *Proceedings of the Association for Information Science and Technology, 61*(1), 103–114. https://doi.org/10.1002/pra2.1012

[C1] Khoury, C., Owen-Smith, A., **Joh, U.**, Duan, Y., & Hemsley, J. (2023). Multi-modal crisis discourse and collective sensemaking on TikTok. *Proceedings of the Association for Information Science and Technology, 60*(1), 203–212. https://doi.org/10.1002/pra2.781

Refereed Conference Posters

[P3] **Joh, U.**, Liu, Q., & Zhang, B. (2026, forthcoming). When AI memory fails: Reframing LLM agent memory as a knowledge organization problem. *Proceedings of the Association for Information Science and Technology*.

[P2] Smith, A. O., **Joh, U.**, & Hemsley, J. (2025). An indexed account of ephemeral culture: Algorithmic contexts of memes. *Proceedings of the Association for Information Science and Technology, 62*(1), 1678–1680. https://doi.org/10.1002/pra2.1506

[P1] Duan, Y., Khoury, C., Smith, A. O., **Joh, U.**, & Hemsley, J. (2023). Comparing crisis communication on TikTok and YouTube: A case study of the 2023 California floods. *Proceedings of the Association for Information Science and Technology, 60*(1), 949–951. https://doi.org/10.1002/pra2.908

Non-Archival Conference Extended Abstracts

[E2] **Joh, U.**, & Introne, J. (2024, July 18–20). *A psychological model of narrative diffusion* [Conference presentation]. 10th International Conference on Computational Social Science, Philadelphia, PA, United States.

[E1] Sun, Q., **Joh, U.**, & Introne, J. (2024, July 18–20). *Empirical validation of a model of narrative diffusion* [Conference presentation]. 10th International Conference on Computational Social Science, Philadelphia, PA, United States.

Manuscripts Under Review

[U3] **Joh, U.**, Stromer-Galley, J., & Li, Y. (2026). *From message similarity to communicative repertoires in coordinated political advertising* [Manuscript under review at the 21st International AAAI Conference on Web and Social Media (ICWSM 2027)].

[U2] **Joh, U.**, & Zhang, B. (2026). *Same personality setting, different assistance? Auditing personality controls in AI assistants* [Manuscript under review at the ACM CHI Conference on Human Factors in Computing Systems (CHI 2027)].

[U1] **Joh, U.**, & Zhang, B. (2026). *Memory Use and personalization in AI assistants* [Working title during blind review; manuscript under review at ACL Rolling Review (ARR), August 2026 cycle].

Teaching Experience

Lab Session Instructor

2023–2026

Syracuse University — Syracuse, NY, USA

Introduction to Applied Data Science (IST 387) — Fall 2023; Spring/Fall 2025; Spring 2026

Led six practice-based lab sections (approximately 25 students each) across four semesters for undergraduates in data analytics and related majors with prior programming experience. Reinforced foundational data science concepts through R exercises and live coding; graded homework, exams, and team projects; held office hours; and received separate teaching evaluations.

Guest Lecturer

2023–2026

Syracuse University — Syracuse, NY, USA

Understanding Large Language Models: Theory and Practice (IST 700)
Instructor: Dr. Bei Yu

Mechanistic Interpretability — Apr 15, 2026

Introduced how mechanistic interpretability can explain internal representations of large language models, using sparse autoencoders as a central example.

Attention in Transformers — Feb 11, 2026

Explained how attention represents contextual relationships, connecting conceptual foundations with a hands-on implementation.

Research Experiences for Undergraduates (REU)

Computational Social Science Research Process — Jun 21, 2024

Introduced the research process from research questions and data collection through analytical design, interpretation, and broader implications.

Influencers in Network Social Media (IST 400)
Instructor: Dr. Yiqi Li

Text Analysis and Topic Modeling — Apr 17, 2023

Introduced computational text analysis and topic modeling for studying social media discourse, with an applied Twitter-data demonstration in R.

Teaching Assistant
2020–2021
Sungkyunkwan University
Seoul, Korea

International Finance (ECO 3015) — Fall 2020; Fall 2021

Labor Economics (ECO 3009) — Spring 2021

Public Finance (ECO 2010) — Spring 2021

Honors and Awards

Summer Dissertation Fellowship, Syracuse University — May 2026
Selected from a competitive pool of more than 70 applicants; awarded $4,500 in support of dissertation research.

Research Excellence Dissertation Grant, Syracuse University — Nov 2025
Selected from a competitive pool (approximately 25% acceptance rate); awarded $2,500 in support of dissertation research.

Top 6 Finalist, SIG-SM Social Media Research Competition, Association for Information Science and Technology — Nov 2025
Selected as a finalist (37.5% acceptance rate) and invited to present at the SIG-SM Social Media Workposium.

Best Full Research Paper Award, iConference — Mar 2025
Co-authored as second author, with a focus on data analysis.

Brain Korea 21 Research Scholarship, National Research Foundation of Korea — Fall 2020–Summer 2022
Merit-based monthly stipend for excellent students from research-focused departments.

Samsung Fellowship, Samsung Foundation — Mar 2012
Merit-based scholarship for excellence in the college entrance examination; full scholarship for four years of undergraduate study.

Professional Service

Peer Reviewer

International AAAI Conference on Web and Social Media (ICWSM)	2025, 2026
Association for Information Science and Technology (ASIS&T)	2023, 2026
ACM Transactions on Social Computing	2026
Hawaii International Conference on System Sciences (HICSS)	2026
Scientific Reports	2026
Discover Global Society	2026
International Conference on Computational Social Science (IC2S2)	2024, 2025

Departmental Service

Personnel Committee, Syracuse University	Fall 2024–Spring 2025
Agenda Committee, Syracuse University	Fall 2023

Skills

Programming	Python, R, SQL, SPARQL, Neo4j, Stata, ArcGIS
Languages	Korean (native), English (fluent)

References

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Dr. Bei Yu
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Dr. Yiqi Li
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